

01 RehabGuide

"A home rehabilitation system empowering office workers with tendonitis to recover safely, precisely, and confidently."

Wearable Design

Innovative Design

This project is a wearable rehabilitation glove designed for tendonitis patients, combined with a complete rehabilitation training service system.

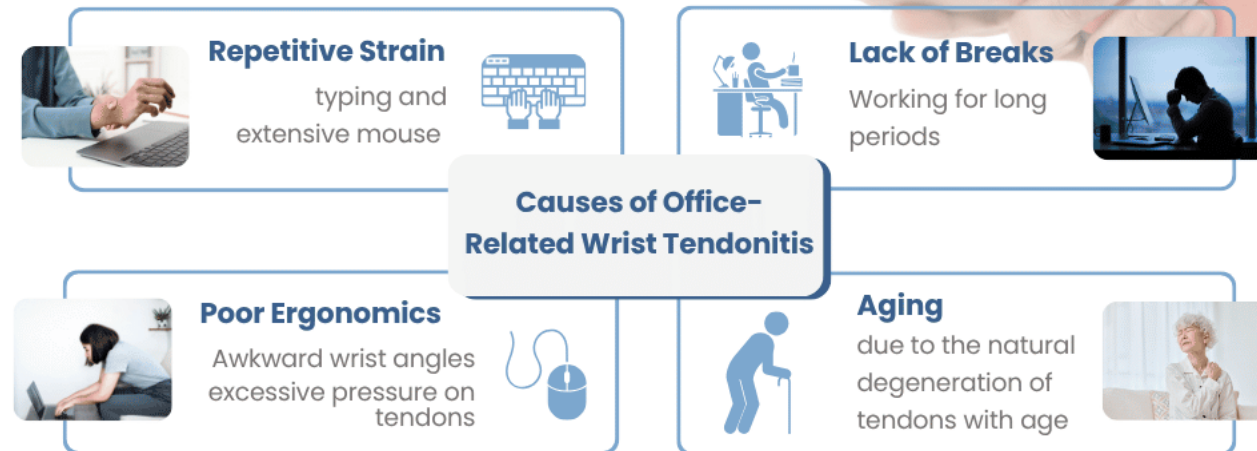
Year: 2025 Spring
Individual Project

BACKGROUND

Does your job require you to spend long hours at a desk, diligently typing away on a keyboard?

What Is Hand and Wrist Tendonitis?

Tendinitis is a common musculoskeletal disorder that affects individuals engaged in repetitive tasks, including computer users, office workers, and manual laborers.



How is wrist tendinitis treated?



Medication



Physical or occupational therapy



Rest



RESEARCH

Who's at Risk for Wrist Tendonitis?

With some studies indicating that over 18% of computer users with repetitive tasks may develop tendinitis.

WHO?

Office workers

The incidence was highest among office workers, indicating a significantly higher risk compared to other job types.

over 18%

Physical therapy or occupational therapy

Wearable systems facilitate rehabilitation outside of traditional clinical settings, allowing patients to exercise comfortably at home, reducing psychological burden.

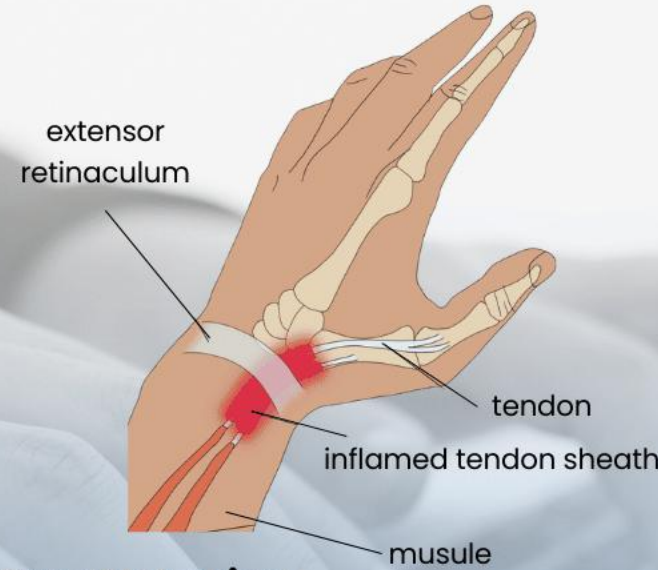


Home rehabilitation

Your physical therapist can create a personalized program of exercises that involves stretching exercises, motion exercises, and eccentric exercises that strengthen the affected muscle and tendon.

WHERE'S HURT?

This sedentary lifestyle can lead to various health issues, such as Hand and Wrist Tendonitis.

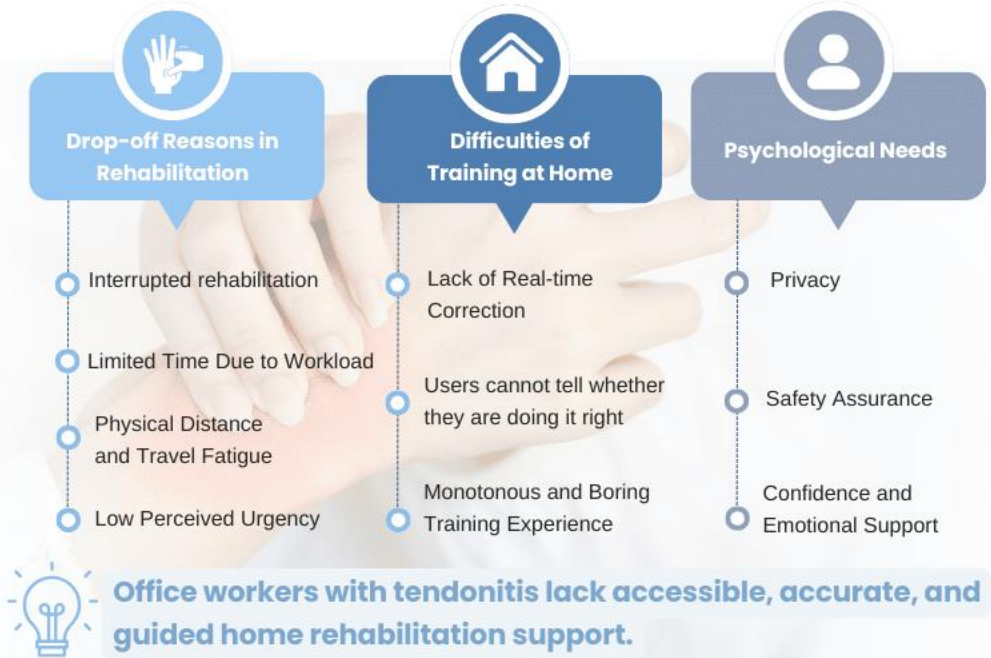


Expert Interview



USER RESEARCH

Around 82% of all workers spend the majority of their working day seated. For the average desk worker, this is often 9 hours or more. 65% of these people develop musculoskeletal problem. Wrist pain affects one in 10 desk workers.



Office workers with tendonitis lack accessible, accurate, and guided home rehabilitation support.

PERSONA



Shawn
Senior Software Engineer

"I wish I had my physiotherapist here to guide me. I just need something to tell me if I'm doing this right or wrong, right now, at home."

PERSONAL INFORMATION

- Age: 25
- Gender: Male
- He was diagnosed with wrist tendonitis.
- Long hours of coding and tight deadlines have taken a toll on his right wrist.

PAINPOINTS

- Chronic Pain & Functional Limitation
- Lack of Guidance & Fear of Making it Worse
- Inconvenient & Time-Consuming Clinic Visits
- No Objective Data

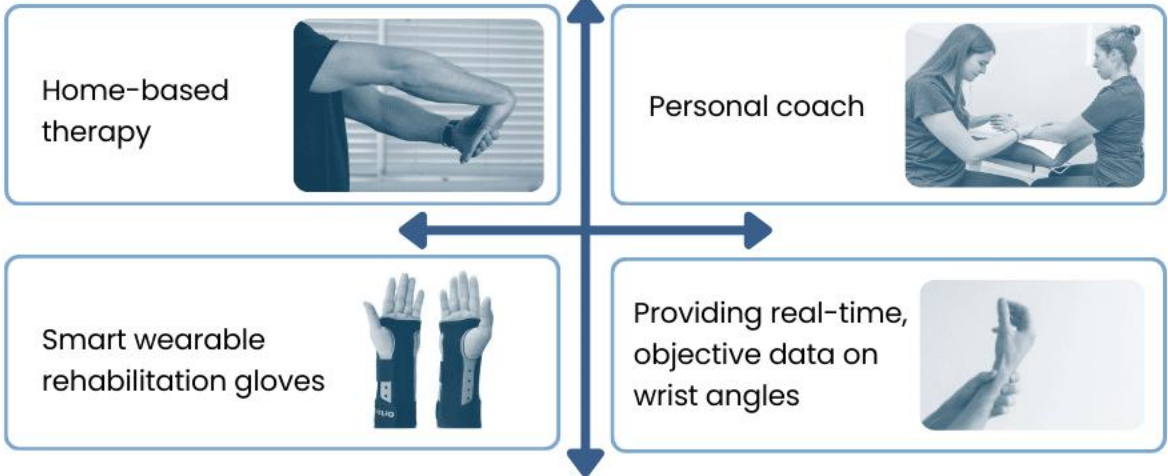
PAIN POINTS

- Incorrect training posture
- Lack of feedback & engagement
- Inconvenient clinic visits
- No objective data

DESIGN GOALS

"Private, Safe, and Correct"

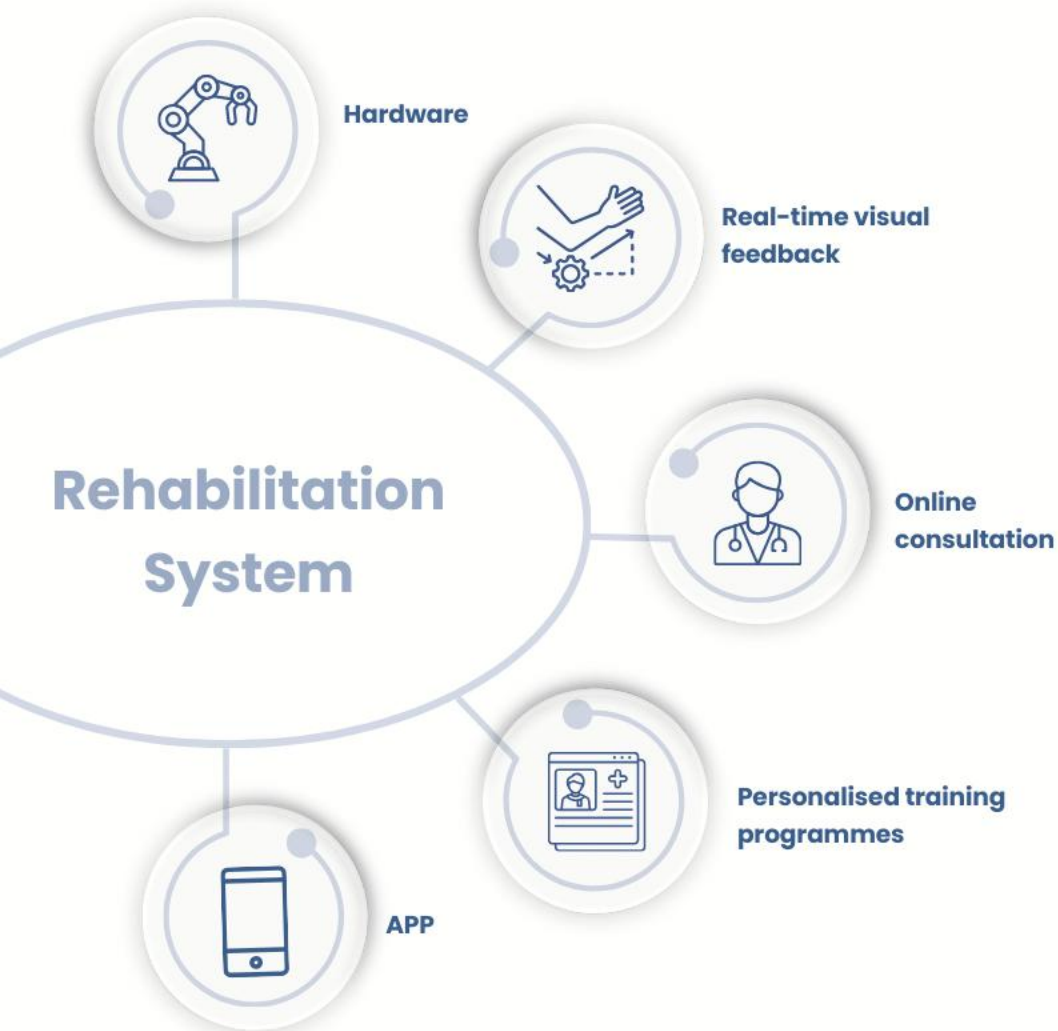
Based on user pain point research, I proposed a set of design goals.



IDEATION

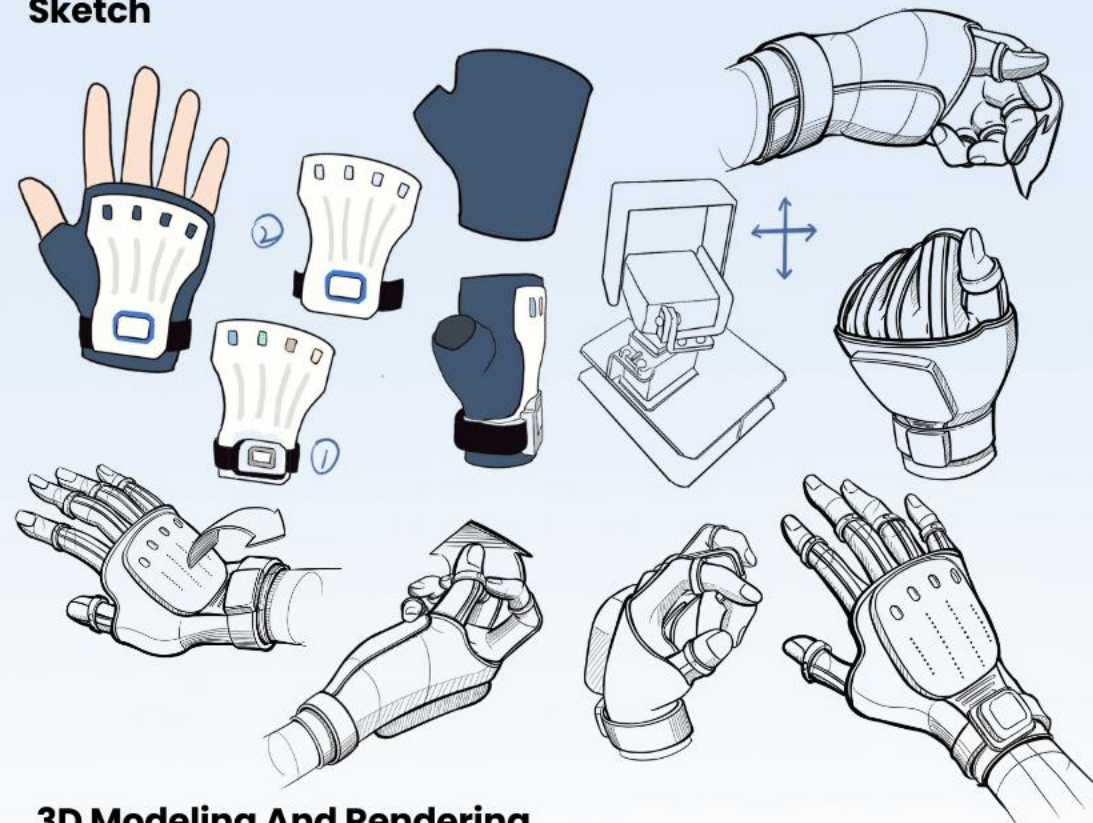


How might we provide physiotherapist-level guidance at home?



PRODUCT DESIGN

Sketch



3D Modeling And Rendering



PRODUCT FUNCTION



Ring sensor

The ring senses the movement of each finger



MPU6050

Sensor Module

Core sensors. Built-in 3-axis gyroscope and 3-axis accelerometer for precise measurement of hand rotation angles and accelerations in three-dimensional space.



LED Progress Bar
(Hand-back light)



Led Light



Finger indicator light



Led Light



Finger indicator light



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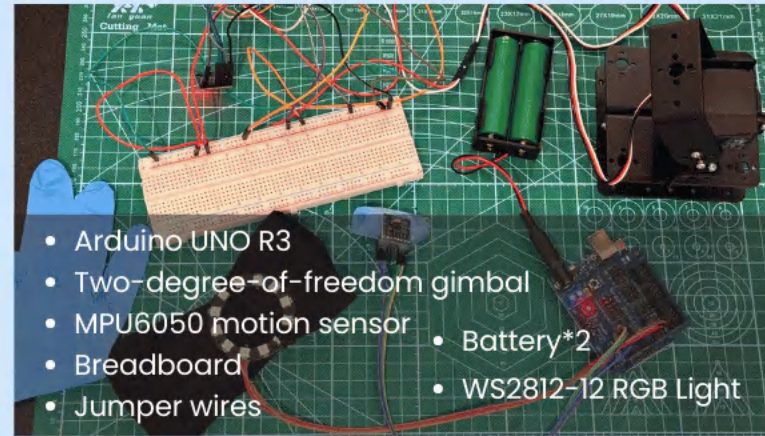


Finger indicator light



TECHNICAL TESTING

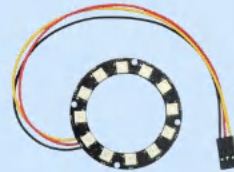
Technique support



Overview of components



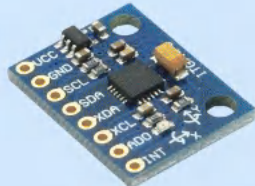
Arduino UNO



WS2812-12 RGB Light



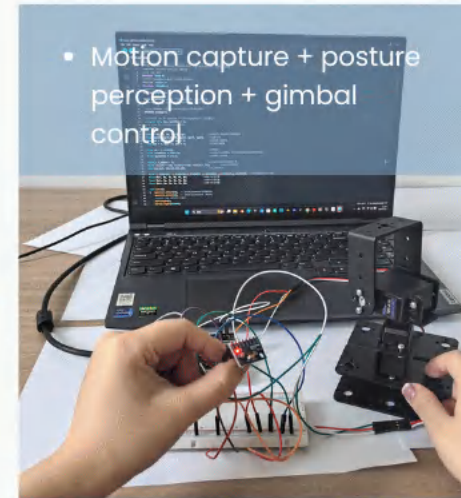
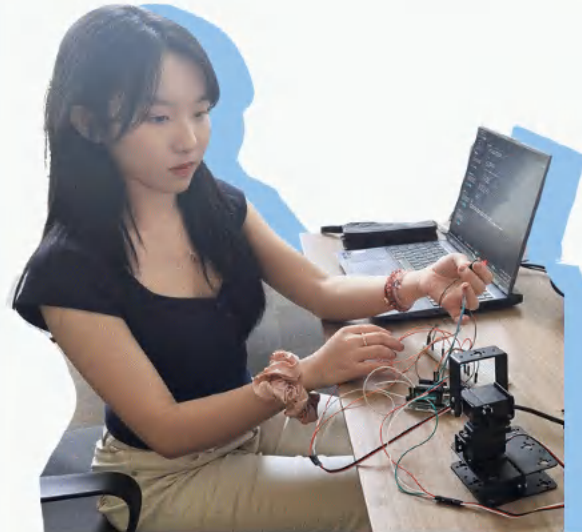
2 DOF Servo Motor Gimbal



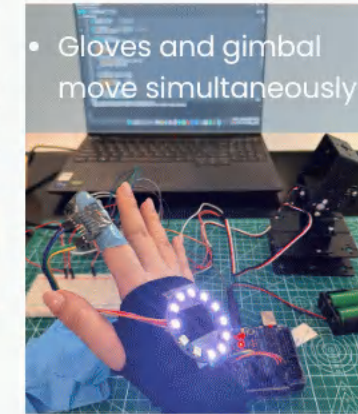
MPU6050 Sensor

PROTOTYPE TESTING

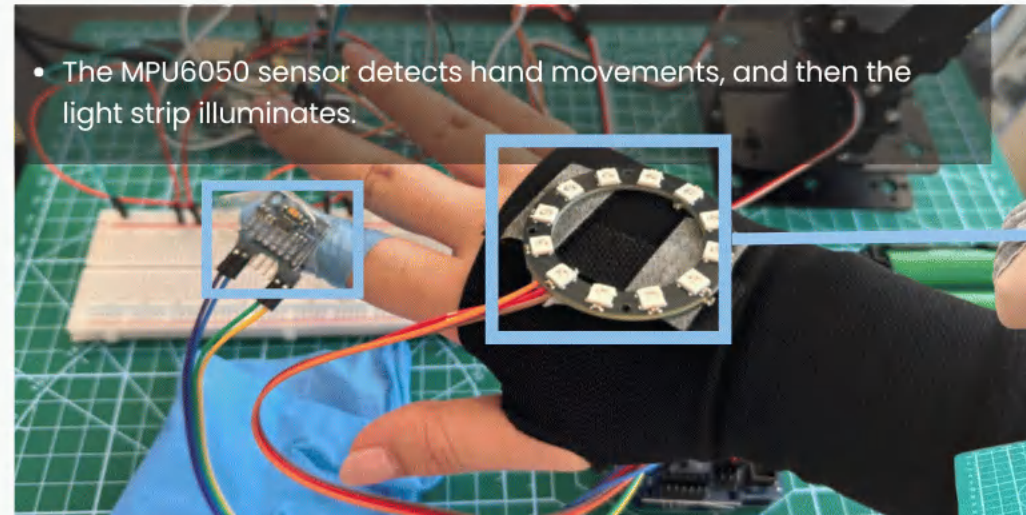
I tried to create the functional points of the gloves.



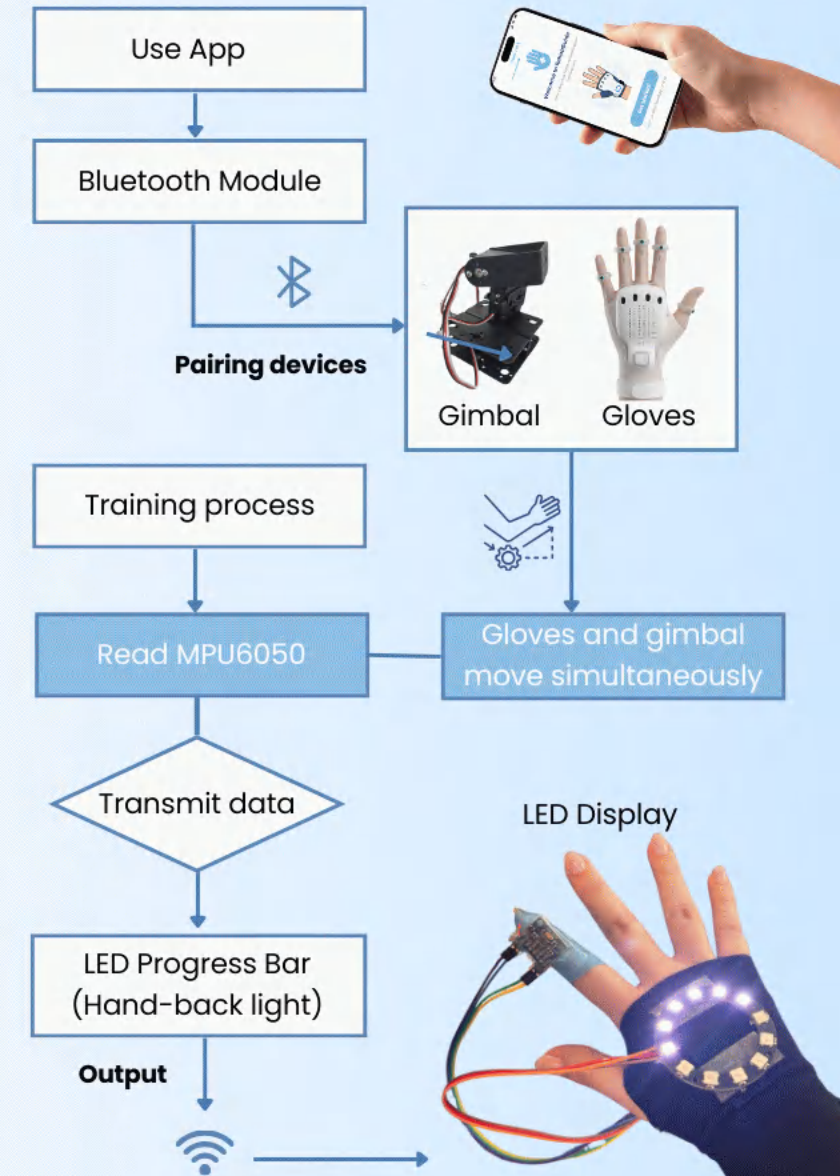
Testing setup



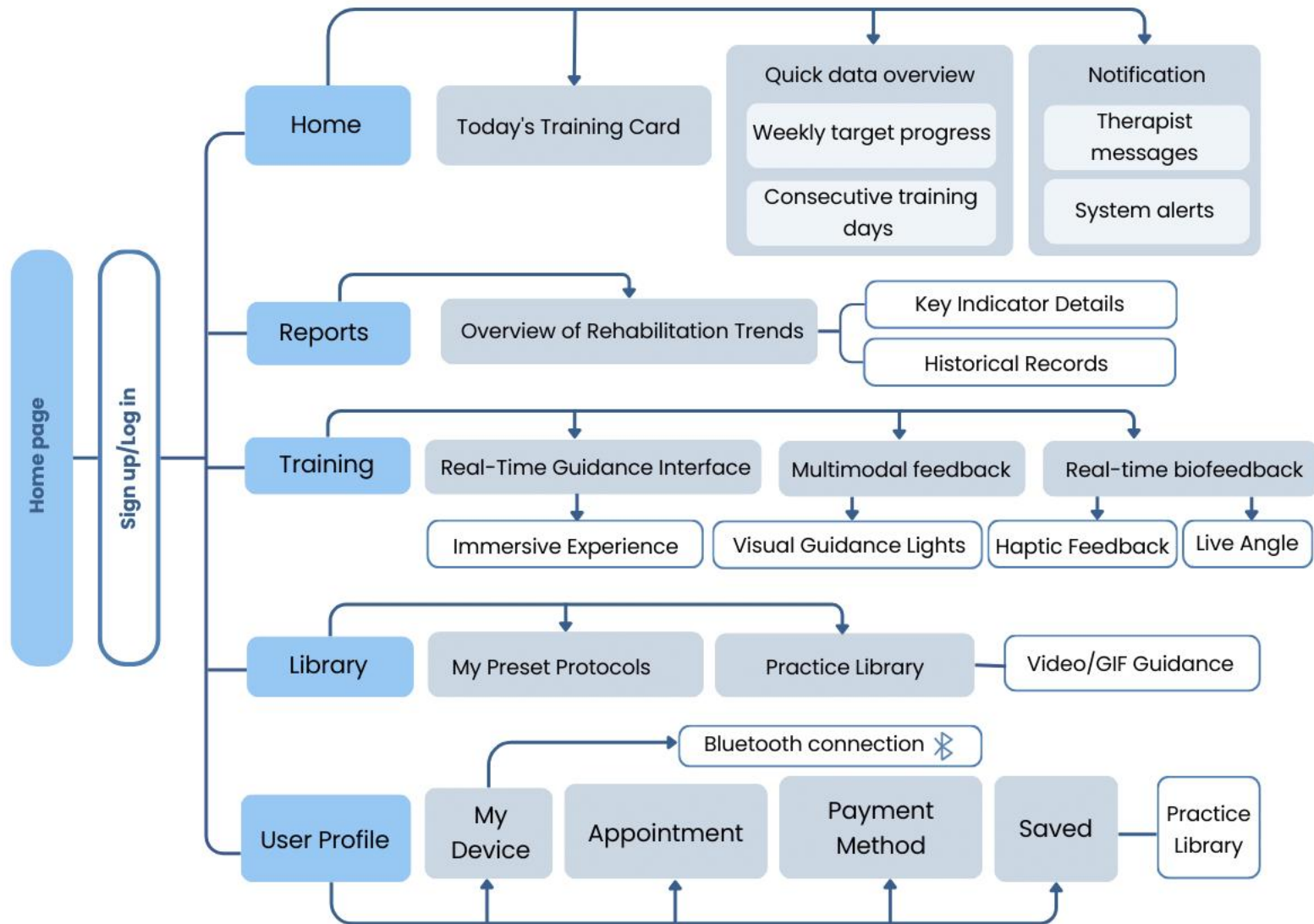
Testing Lighting



WORKING MECHANISM



INFORMATION ARCHITECTURE



APP DEVELOPMENT

Hi-Fi interface



Color



Logo



APP Icon

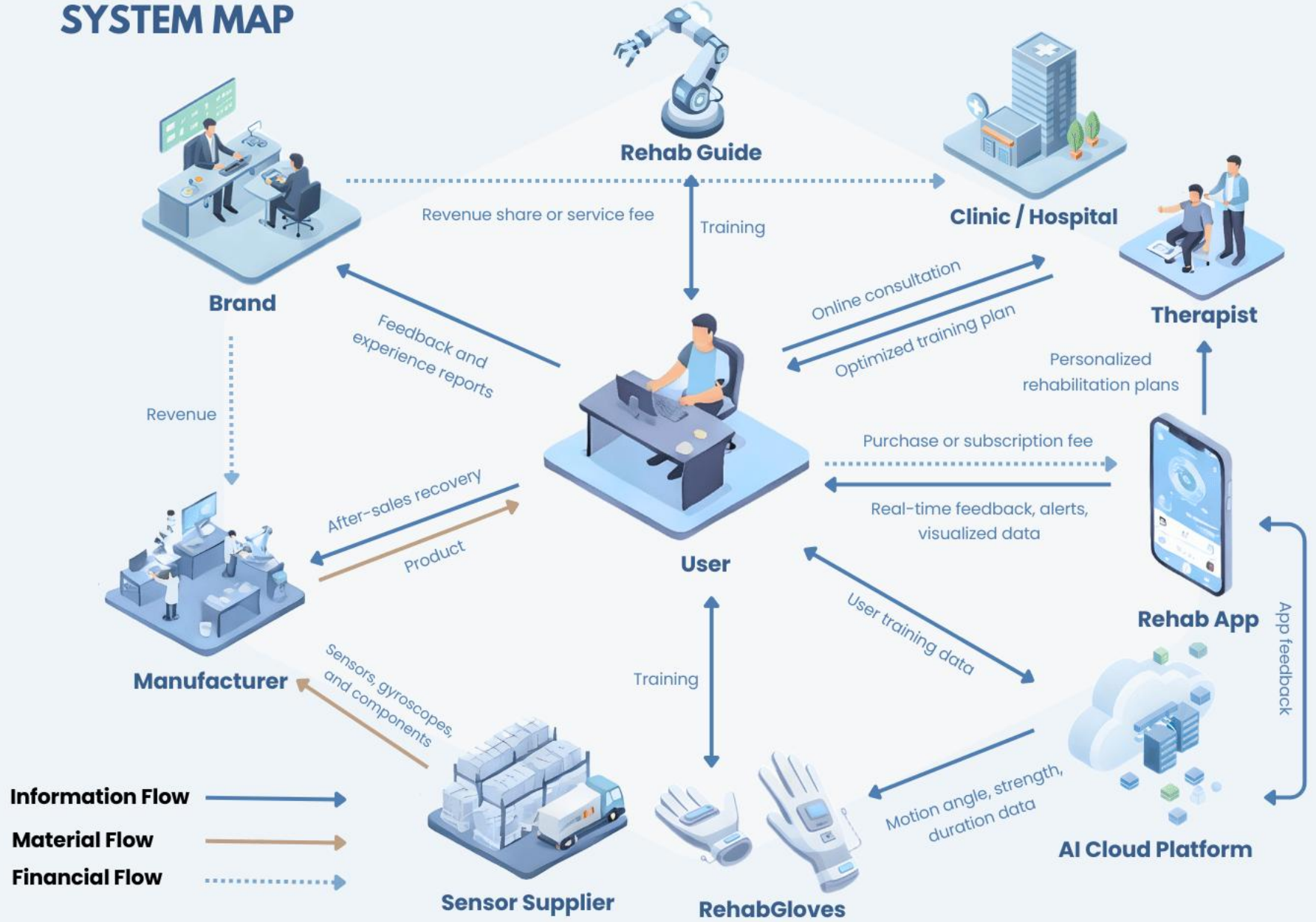
Font



Poppins Bold
Poppins SemiBold
Poppins Medium
Poppins Regular

"Private, Safe, and Correct"

SYSTEM MAP



SERVICE BLUEPRINT

